

TikTok Video Technical Specifications for PreflightQC

(Offline Deliverable QA) — As of August 2026

TL;DR

- The single most authoritative, deterministically-checkable spec set is TikTok's **Content Posting API "Video restrictions"** table (developers.tiktok.com, © 2026), quoted verbatim: *"Supported media formats: MP4 (recommended) WebM MOV"; "Supported codecs: H.264 (recommended) H.265 VP8 VP9"; "Framerate restrictions: Minimum of 23 FPS Maximum of 60 FPS"; "Picture size restrictions: Minimum of 360 pixels for both height and width Maximum of 4096 pixels for both height and width"; "Size restrictions: Maximum of 4GB."* Duration via that endpoint is capped at 10 minutes. These are the only codec/framerate/pixel bounds TikTok states as hard machine constraints and should form the backbone of PreflightQC's normal-upload profile.
- For ads, the authoritative source is the TikTok Business Help Center (ads.tiktok.com/help). In-Feed Auction Non-Spark ads carry concrete, enforceable-looking limits — verbatim *"File size: Less than or equal to 500 MB. Bitrate: More than or equal to 516 kbps"*, min dimensions by ratio, containers .mp4/.mov/.mpeg/.3gp/.avi, ≤10 min. TopView/Reservation tighten this to ≥2,500 kbps and 5–60 s. Spark Ads (Pull) inherit the organic file and have "no restrictions."
- Almost everything at the fine-grained codec level (profile/level, chroma subsampling, bit depth, HDR, faststart/moov, PAR, GOP, LUMS) is **UNDOCUMENTED** by TikTok. PreflightQC must treat those as WARN/INFORMATIONAL, never hard-fail, because TikTok re-encodes every upload server-side and publishes no rejection thresholds for them.

Key Findings

1. **Two distinct authoritative regimes exist and must not be conflated.** (a) Organic/creator uploads are governed by the consumer Help Center (support.tiktok.com) and, for programmatic upload, the Content Posting API media guide (developers.tiktok.com). (b) Paid ads are governed by the Business Help Center (ads.tiktok.com/help), with different numeric limits per placement. A rule that is real for one regime is not automatically valid for the other.
2. **The Content Posting API is the richest machine-readable spec TikTok publishes.** It is the only official TikTok source that enumerates codecs, frame-rate bounds, and pixel min/max. Because the app upload paths (in-app, web/TikTok Studio) share TikTok's ingestion backend, these bounds are the best available proxy for organic-upload hard limits — but TikTok does not formally state the API table governs in-app uploads, so that inference must be labelled as such.
3. **TikTok's own official sources contradict each other on maximum organic duration and file size** (detailed in §7): Help Center "Camera tools" (record ≤10 min in-app / upload ≤60 min), Help Center "Tools for creators"/TikTok Studio (≤30 min, <10 GB), Content Posting API (≤10 min, ≤4 GB), and Creator Academy marketing (60 min, 30 GB).  This is a genuine documentation inconsistency,

not a research gap, and PreflightQC must resolve it by making the target *upload path* an explicit profile the user selects.

4. **Very little is a true HARD REQUIREMENT.** Most numbers are “supported ranges,” minimums, or “recommended.” TikTok re-encodes/transcodes all uploads, so many parameters are silently normalized rather than rejected. Conservative QA should hard-fail only on the handful of unambiguous constraints and warn on everything else.
 5. **HDR, bit depth, colour space, chroma subsampling, faststart/moov, PAR/anamorphic, GOP, interlacing, and loudness are entirely undocumented by TikTok as pass/fail criteria.** TikTok did publicly enable 10-bit HDR *capture* on the Pixel 7 in 2022 (Google, at its October 6, 2022 Pixel 7 launch, stated it had “partnered with Snap, TikTok, and YouTube so your HDR videos look vibrant when shared on your favorite apps”; Snapchat and TikTok were the first Android apps to support 10-bit HDR video on the Pixel 7 series), but TikTok publishes no HDR *delivery* spec or restriction. These must be INFORMATIONAL only.
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1. SOURCE REGISTER

Tier 1 — Official TikTok consumer Help Center (support.tiktok.com / tiktok.com)

- **“Tools for creators” (TikTok Studio)** — support.tiktok.com/en/using-tiktok/creating-videos/creator-tools-on-tiktok. Verbatim Studio web-upload requirements: *“Uploaded videos must be: In MP4 or WebM file format; 720x1280 resolution or higher; Up to 30 minutes in length; Less than 10 GB.”* Access: 9 Aug 2026 (direct fetch). Used for: organic web/Studio upload matrix.
- **“Camera tools”** — support.tiktok.com/en/using-tiktok/creating-videos/camera-tools. Verbatim: *“Videos you record in TikTok can be up to 10 minutes long. Videos you upload in TikTok can be up to 60 minutes long.”* Plus: *“If you select a sound before you record or upload, the available video length may be determined by the sound length.”* Access: 9 Aug 2026. Used for: in-app duration limits. (Note: English page authoritative; non-English mirrors lag.)
- **“Creating videos” / “Creator Center” index** — support.tiktok.com/en/using-tiktok/creating-videos. Confirms article taxonomy (Camera tools, Tools for creators, TikTok Studio, Series, Photo mode). Access: 9 Aug 2026.

Tier 2 — TikTok Business / Ads (ads.tiktok.com/help)

- **“TikTok Auction In-Feed Ads”** — ads.tiktok.com/help/article/tiktok-auction-in-feed-ads. Last updated June 2026.  Non-Spark + Spark (Pull). Access: 9 Aug 2026. Primary ad source.
- **“TopView ad specifications”** — ads.tiktok.com/help/article/tiktok-reservation-topview. Last updated June 2026. Access: 9 Aug 2026.
- **“TikTok reservation in-feed ad specifications (incl. Reach & Frequency)”** — ads.tiktok.com/help/article/tiktok-reservation-in-feed-ads-reach-frequency. Access: 9 Aug 2026.
- **“Global App Bundle video ad specifications”** — ads.tiktok.com/help/article/global-app-bundle-video-ad-specifications. Last updated July 2025. Access: 9 Aug 2026.

- **"Specifications for TikTok Pangle ad assets"** — ads.tiktok.com/help/article/specifications-for-pangle-ad-assets. Last updated July 2025. Access: 9 Aug 2026.
- **"About Carousel Ads" / "Specifications for Carousel Ads"** — ads.tiktok.com/help/article/carousel-ads-specifications-for-carousel-ads. Access: 9 Aug 2026.
- **"About Spark Ads" / "Spark Ads in Video Shopping Ads"** — ads.tiktok.com/help/article/spark-ads-spark-ads-video-shopping-ads. Access: 9 Aug 2026.
- **"Creative Specs for Streaming Ads" / "Creative specs for Automotive Ads" / "Specifications for Video Packages"** — ads.tiktok.com/help/article/... (catalog/auto-generated video). Access: 9 Aug 2026.

Tier 3 — TikTok Developer (developers.tiktok.com)

- **"Media Transfer Guide" → Video restrictions table** — developers.tiktok.com/doc/content-posting-api-media-transfer-guide. Core codec/framerate/pixel/size table (quoted in TL;DR). Also chunk rules, verbatim: *"Each chunk must be at least 5 MB but no greater than 64 MB, except for the final chunk, which can be greater than chunk_size (up to 128 MB)... There must be a minimum of 1 chunk and a maximum of 1000 chunks."* Access: 9 Aug 2026. Primary organic-programmatic source.
- **"Get Started - Direct Post" / "Get Started - Upload"** — [developers.tiktok.com/doc/content-posting-api-get-started\(-upload-content\)](https://developers.tiktok.com/doc/content-posting-api-get-started(-upload-content)). "MP4 + H.264" example;  Query Creator Info returns per-creator `max_video_post_duration_sec` (e.g., 300). Access: 9 Aug 2026.
- Duration note, verbatim from Media Transfer Guide: *"All TikTok creators can post 3-minute videos, while some have access to post 5-minute or 10-minute videos. The longest video a developer can send via the initialize Upload Video endpoint is 10 minutes."*

Corroborating press (leads, not spec authority): TechCrunch, Feb 28 2022 — *"TikTok expands max video length to 10 minutes, up from 3 minutes,"* quoting a TikTok spokesperson: *"we're excited to start rolling out the ability to upload videos that are up to 10 minutes."* Notebookcheck/XDA, Oct 2022 — Pixel 7 10-bit HDR.

Third-party (LEADS ONLY — never promoted to authority): Renderforest, PostFast, StackInfluence, Riverside, Descript, Aiarty, Insense, Adnabu, Triple Whale, Recharm, TikAdSuite, Sovran, soona, AdRate, Moda. Used only to surface candidate rules subsequently checked against official sources.

2. NORMAL UPLOAD SPECIFICATION MATRIX

Property	Value	Source (tier)	Classification	Confidence	Locally measurable?
Container (web/Studio)	MP4 or WebM	Help Center "Tools for creators" (T1)	DOCUMENTED LIMIT	High	Yes (ftype brand)

Container (Content Posting API)	MP4 (recommended), WebM, MOV	Media Transfer Guide (T3)	DOCUMENTED LIMIT	High	Yes
Container (in-app, exhaustive list)	Not officially enumerated (MP4/MOV widely accepted)	Third-party leads	UNKNOWN	Low	Yes
Video codec	H.264 (recommended), H.265, VP8, VP9 tiktok	Media Transfer Guide (T3)	DOCUMENTED LIMIT	High	Yes
Codec profile / level	Not documented	—	UNKNOWN	Low	Yes
Audio codec	AAC recommended (H.264+AAC example); not stated as hard rule	Get Started (T3)	RECOMMENDATION	Medium	Yes
Min resolution	720×1280 or higher (Studio); Renderforest 360 px min per side (API) tiktok	Help Center + API (T1/T3)	DOCUMENTED LIMIT	High	Yes
Max resolution	4096 px max per side (API); tiktok output capped ~1080p after transcode	API (T3)	DOCUMENTED LIMIT	High (input)	Yes (input only)
Aspect ratio	9:16 native/recommended; 1:1 & 16:9 accepted (letterboxed)	Help Center + leads	RECOMMENDATION	Medium	Yes
Orientation	Vertical/horizontal/square accepted	Help Center	RECOMMENDATION	Medium	Yes
Frame rate	23–60 FPS	Media Transfer Guide (T3)	DOCUMENTED LIMIT	High	Yes (nominal)
VFR vs CFR	Not documented	—	UNKNOWN	Low	Yes
Duration min	3 s commonly cited; NOT in primary TikTok spec	Third-party leads	UNKNOWN	Low	Yes

Duration max	10 min in-app record; 60 min in-app upload; PostFast 30 min Studio; Renderforest 10 min API tiktok	Help Center + API (T1/T3)	DOCUMENTED LIMIT (contradictory)	High but conflicting	Yes
File size max	<10 GB (Studio); 4 GB (API); tiktok ~72 MB Android / ~287.6 MB iOS (app, third-party) Aiarty	Help Center + API (T1/T3); leads (mobile)	DOCUMENTED LIMIT (Studio/API); UNKNOWN (mobile)	High (Studio/API), Low (mobile)	Yes
Audio required	Not mandatory for organic; silent accepted	Third-party leads	UNKNOWN (not officially stated)	Low	Yes (track presence)
Sample rate / channels	Not documented	—	UNKNOWN	Low	Yes
Bitrate (video/audio)	No organic min/max documented	—	UNKNOWN	Low	Yes
Colour / HDR / bit depth / chroma	Not documented; 10-bit HDR capture enabled Pixel 7 (2022)	Press (leads)	UNKNOWN	Low	Yes (tags)
Faststart / moov / PAR / GOP / interlacing	Not documented	—	UNKNOWN	Low	Yes

3. AD SPECIFICATION MATRIX (by placement)

In-Feed Auction — Non-Spark (updated June 2026): Dimensions 9:16 $\geq 540 \times 960$ / 16:9 $\geq 960 \times 540$ / 1:1 $\geq 640 \times 640$; containers .mp4/.mov/.mpeg/.3gp/.avi; duration up to 10 min; **file size ≤ 500 MB; bitrate ≥ 516 kbps** (verbatim from the Auction In-Feed page). → These read as HARD/DOCUMENTED LIMITS enforced by Ads Manager.

In-Feed Auction — Spark Ads (Pull): container .mp4 or .mov; **video duration: “No restrictions”** (inherits the organic post; Spark-eligible organic videos are capped at 10 min). → No independent hard file limits beyond the organic upload path.

TopView / Reservation (Non-Spark & Spark Push) (updated June 2026): 9:16 $\geq 540 \times 960$; .mp4/.mov/.mpeg/.3gp; duration 5–60 s (recommend 9–15 s); Triple Whale ≤ 500 MB; **bitrate $\geq 2,500$ kbps**; audio “*must not be completely silent*” tiktok (creative policy, not a file-metadata rule). Spark Pull: no duration restriction. tiktok

Reservation In-Feed (Reach & Frequency): same ratio minimums; 5–60 s; ≤ 500 MB; $\geq 2,500$ kbps. The

official page emphasises creative-policy rules (no watermarks including TikTok's, must not mimic the TikTok interface). Treat numeric limits as DOCUMENTED LIMIT.

Global App Bundle (CapCut/Fizzo) (updated July 2025): 9:16 $\geq 720 \times 1280$ / 16:9 $\geq 1280 \times 720$ / 1:1 $\geq 640 \times 640$; .mp4/.mov/.mpeg/.avi; 5–60 s (recommend 21–30 s); ≤ 500 MB; **bitrate ≥ 516 kbps recommended** (note: "recommended," not "required").

Pangle (updated July 2025): video up to 10 min; resolutions 1280×720 / 720×1280 / 720×720; .mp4/.mov/.mpeg/.avi; ≤ 500 MB. Playable = .zip ≤ 5 MB.

Carousel Ads: image-only (2–35 images, JPG/PNG); SocialBee no video component in Standard Carousel. VSA (Video Shopping Ads) Carousel is catalog-driven. Not a video deliverable for local video QA except to flag that a video sent to a carousel path is mis-targeted.

Streaming Ads / Catalog Video Packages / Automotive: 3x4 recommended for streaming; TikTok catalog video via hosted `video_link`; opening/end scenes MP4/MOV(/FLV/MKV/WebM) 1–10 s. TikTok Niche; DOCUMENTED but placement-specific.

Common ad codec note: TikTok ad help pages specify *containers* but do **not** authoritatively enumerate *codecs*. H.264+AAC is a strong recommendation from many secondary sources, not an official ad-level hard rule; classify as RECOMMENDATION for ad profiles.

4. DETERMINISTIC FAIL RULES (safe hard-fails from local inspection)

Scope every rule to the profile the customer selects; do not apply ad limits to organic files or vice versa.

Normal upload — Content Posting API profile (safest proxy for programmatic delivery):

- Container $\notin \{\text{MP4, MOV, WebM}\} \rightarrow \text{FAIL}$.
- Video codec $\notin \{\text{H.264, H.265, VP8, VP9}\} \rightarrow \text{FAIL}$.
- Any pixel dimension < 360 or $> 4096 \rightarrow \text{FAIL}$.
- Frame rate (nominal/average) < 23 or > 60 FPS $\rightarrow \text{FAIL}$ (see VFR caveat).
- Duration > 10 min $\rightarrow \text{FAIL}$; file size > 4 GB $\rightarrow \text{FAIL}$ — **only** when targeting the API path.

Normal upload — Studio/web profile: container $\notin \{\text{MP4, WebM}\} \rightarrow \text{FAIL}$; any dimension below 720×1280 $\rightarrow \text{FAIL}$; duration > 30 min $\rightarrow \text{FAIL}$; file size ≥ 10 GB $\rightarrow \text{FAIL}$.

In-Feed Auction Non-Spark ad profile:

- File size > 500 MB $\rightarrow \text{FAIL}$. tiktok
- Video bitrate < 516 kbps $\rightarrow \text{FAIL}$. tiktok
- Container $\notin \{\text{.mp4, .mov, .mpeg, .3gp, .avi}\} \rightarrow \text{FAIL}$.
- Dimensions below ratio minimum (9:16 $< 540 \times 960$, 16:9 $< 960 \times 540$, 1:1 $< 640 \times 640$) $\rightarrow \text{FAIL}$.
- Duration > 10 min $\rightarrow \text{FAIL}$.

TopView / Reservation ad profile: duration outside 5–60 s → FAIL; bitrate < 2,500 kbps → FAIL; file size > 500 MB → FAIL; 9:16 below 540×960 → FAIL.

5. WARN-ONLY RULES (recommendations / documented-but-unconfirmed)

- Aspect ratio ≠ 9:16 → WARN (recommendation; 1:1/16:9 accepted, letterboxed).
 - Resolution < 1080×1920 → WARN (1080p recommended; 720×1280 is the documented Studio minimum).
 - Audio codec ≠ AAC → WARN (recommendation; TikTok transcodes).
 - Missing audio track → WARN (organic accepted silent; ads carry a “not completely silent” creative policy, not a file-metadata hard rule).
 - Mobile file-size caps (~72 MB Android / ~287.6 MB iOS) Aiarty → WARN, only when targeting mobile-app upload; these are third-party figures absent from primary docs.
 - Duration 30–60 min → WARN (Studio 30 min vs in-app 60 min ambiguity).
 - Video bitrate below ~2,000–2,500 kbps for 1080p organic → WARN (quality guidance).
 - Frame rate not CFR (i.e., VFR detected) → WARN (may be reinterpreted on transcode).
 - GAB “≥516 kbps” → WARN not FAIL (page says “recommended”).
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6. INFORMATIONAL METADATA (report, do not judge)

Colour primaries / transfer characteristics / matrix; HDR type (HDR10/HLG/Dolby Vision); bit depth (8/10-bit); chroma subsampling (4:2:0/4:2:2); pixel aspect ratio / anamorphic flags; rotation metadata; faststart / moov atom position; GOP / keyframe interval; interlaced vs progressive; count of video/audio/subtitle/data/timcode tracks; alpha channel; integrated loudness (LUFS) / true peak; audio sample rate & channel count; container brand/minor version. TikTok documents none of these as pass/fail; surface values for operator insight only. **Safe zones are explicitly a layout concern, not a file-metadata property** — PreflightQC cannot and should not judge safe-zone compliance from container metadata.

7. UNRESOLVED SPECIFICATIONS (contradictions & gaps)

Contradiction A — Max organic duration: Camera tools = 60 min (upload) / 10 min (record); Tools for creators/Studio = 30 min; tiktok Content Posting API = 10 min; Creator Academy marketing = 60 min.

TikTok Four official numbers for overlapping paths.

Contradiction B — Max organic file size: Studio = <10 GB; tiktok Renderforest Content Posting API =

≤4 GB; Creator Academy marketing = 30 GB; TikTok third-party mobile app = ~72/287.6 MB.

StackInfluence

Aiarty

No single reconciled figure.

Undocumented by TikTok (all UNKNOWN): exhaustive in-app container list; codec profile/level constraints; audio codec/sample-rate/channel hard rules; minimum duration in primary docs; minimum/maximum bitrate for organic; HDR/colour/bit-depth/chroma acceptance; faststart requirement; PAR/anamorphic handling; GOP/keyframe rules; interlacing acceptance; loudness normalization targets; VFR vs CFR policy; whether the API “Video restrictions” table governs in-app uploads; whether ad codecs beyond container are enforced.

8. INSPECTION-ENGINE REQUIREMENTS (MediaInfo/ffprobe field mapping)

- **Container/brand:** `ffprobe format.format_name , format.tags.major_brand ; MediaInfo Format , CodecID .`
 - **Video codec/profile/level:** `ffprobe stream.codec_name , profile , level .`
 - **Dimensions + rotation:** `width , height , plus stream.tags.rotate / displaymatrix` side-data to compute *displayed* orientation and AR.
 - **Aspect ratios / anamorphic:** `sample_aspect_ratio , display_aspect_ratio .`
 - **Frame rate / VFR:** `r_frame_rate , avg_frame_rate ;` VFR via packet-PTS analysis.
 - **Duration:** `format.duration / stream.duration .`
 - **File size:** `format.size .`
 - **Video bitrate:** `stream.bit_rate` or `format.bit_rate` (fallback `size×8/duration`).
 - **Audio:** `codec_name , sample_rate , channels , channel_layout , audio_bit_rate ;` presence/count of audio streams.
 - **Colour/HDR:** `color_primaries , color_transfer , color_space , bits_per_raw_sample / pix_fmt` (bit depth & chroma), mastering-display/CLL side-data.
 - **Faststart/moov:** `MediaInfo IsStreamable ;` or `ffprobe moov-vs-mdat` order heuristic.
 - **GOP/keyframes:** keyframe interval via frame flags.
 - **Track inventory:** stream counts by type.
 - **Cannot measure locally:** perceptual quality, watermark presence, safe-zone compliance, content-policy/music licensing, server-side re-encode outcome, and per-account `max_video_post_duration_sec` (server-returned, e.g., 300 s).
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9. SUGGESTED MACHINE-READABLE REPRESENTATION (schema

sketch, research artefact only)

```
profile: tiktok_content_posting_api    # or: studio_web_upload, in_feed_auction_nonspark, topv
source_of_truth:
  url: https://developers.tiktok.com/doc/content-posting-api-media-transfer-guide
  tier: developer
  last_verified: 2026-08-09
rules:
- id: container
  field: format.format_name
  op: in
  values: [mp4, mov, webm]
  severity: fail
  classification: documented_limit
- id: video_codec
  field: stream.video.codec_name
  op: in
  values: [h264, hevc, vp8, vp9]
  severity: fail
- id: pixel_bounds
  field: [stream.video.width, stream.video.height]
  op: within
  min: 360
  max: 4096
  severity: fail
- id: framerate
  field: stream.video.avg_frame_rate
  op: within
  min: 23
  max: 60
  severity: fail
- id: duration_max
  field: format.duration_sec
  op: lte
  value: 600
  severity: fail
  applies_when: profile == tiktok_content_posting_api
- id: filesize_max
  field: format.size_bytes
  op: lte
  value: 4294967296
  severity: fail
- id: aspect_ratio
  field: computed.display_aspect_ratio
  op: equals
  value: "9:16"
  severity: warn
  classification: recommendation
informational_fields:
- color_transfer
- color_primaries
- bits_per_raw_sample
```

- pix_fmt
- is_faststart
- gop_size
- audio.sample_rate
- audio.channels
- integrated_loudness_lufs

10. ENGINEERING CAVEATS

- **Spec drift:** ad pages carry “Last updated” dates (In-Feed & TopView June 2026; tiktok tiktok GAB/Pangle July 2025). tiktok tiktok Re-scrape on a schedule; treat any cached rule older than ~6 months as suspect and monitor the developer Changelog for silent edits to the Video restrictions table.
- **Client vs server:** TikTok re-encodes every upload (the 10-minute upload capability itself was a 2022 rollout, per TikTok’s own spokesperson via TechCrunch). Many properties are silently normalized, not rejected — over-strict local failing will reject files TikTok would happily ingest.
- **Regional/localization variation:** non-English Help Center pages lag (the Russian “Camera tools” page still showed old 60 s/3 min limits). tiktok Anchor all rules to English pages.
- **Account/campaign state is out of scope for local QA:** per-creator `max_video_post_duration_sec`, monetization eligibility, and Spark authorization cannot be verified from a file — treat as caveats, never fails.
- **VFR/rotation traps:** evaluate *displayed* dimensions after applying rotation metadata; test frame-rate bounds against nominal/average FPS and flag VFR separately.
- **“Supported formats include...” ambiguity:** TikTok’s wording (“include,” “recommended,” “should”) rarely states rejection. Only Ads Manager numeric limits and the API restriction table are safe to enforce as hard fails; everything hedged is WARN.

11. SOURCE PACK STATUS

The source pack is strong on Tier 2 (ads) and Tier 3 (developer API) and now includes **verbatim Tier 1 Help Center confirmation** for organic Studio/in-app limits (MP4/WebM, 720×1280+, 30 min, <10 GB for Studio; 10-min record / 60-min upload for in-app). It is authoritative for: ad placement limits, API codec/framerate/pixel/size bounds, and organic duration/size (with the four-way contradiction documented rather than papered over). It is weak/silent — mirroring TikTok’s own silence — on: codec profile/level, audio hard rules, HDR/colour/bit-depth/chroma, faststart, PAR, GOP, loudness, and whether the API table governs in-app uploads. Reaching full confidence would require: (a) direct scraping of every ads.tiktok.com placement page for any codec statements; (b) empirical black-box upload testing to derive de-facto rejection thresholds across the in-app, Studio, and API paths; and (c) continuous monitoring of the developer Changelog. No official TikTok source fills the codec-profile/HDR/faststart gaps; they must remain UNKNOWN rather than be populated from third-party blogs.

Recommendations

1. **Ship a profile-based rule engine, not one global ruleset.** Make the user pick the target path (In-app upload / TikTok Studio web / Content Posting API / In-Feed Auction / TopView-Reservation / Global App Bundle / Pangle). Each profile draws from its own authoritative page. This is the single most important design decision because TikTok's numbers genuinely differ by path.
2. **Hard-fail only on §4.** For the default organic profile, use the Content Posting API table (MP4/MOV/WebM; H.264/H.265/VP8/VP9; 360–4096 px; 23–60 FPS; ≤10 min; ≤4 GB). For ads, use the placement-specific numeric limits (≤500 MB and ≥516 kbps for In-Feed Auction; ≥2,500 kbps and 5–60 s for TopView).
3. **Everything hedged goes to WARN (§5), everything undocumented to INFORMATIONAL (§6).** Never fail a file for aspect ratio, resolution below 1080p, missing AAC, HDR, faststart, or loudness.
4. **Surface the duration/size contradiction to the user** rather than silently choosing one limit: if a 45-minute file is checked against the "organic" profile, warn that it exceeds Studio's 30-min and the API's 10-min limits but is within Camera-tools' 60-min in-app-upload figure.
5. **Benchmarks that would change these rules:** (a) a change to the developer "Video restrictions" table → update the organic hard-fail bounds; (b) a new "Last updated" date on any ads page → re-verify that placement's numbers; (c) publication by TikTok of any codec-profile, HDR, or faststart requirement → promote the relevant INFORMATIONAL field to WARN or FAIL; (d) reconciliation of the duration/size figures on the Help Center → collapse the multi-value duration/size rows into one.

Caveats

- Figures marked "third-party" (mobile app 72 MB / 287.6 MB caps; 3-second minimum duration; ad-level codec enforcement) are **not** confirmed by primary TikTok documentation and are included only as leads; do not hard-fail on them.
- TikTok transcodes all uploads, so a file that violates a *recommendation* will typically still publish (often with quality loss), which is precisely why PreflightQC must separate rejection conditions from quality guidance.
- The Content Posting API path additionally imposes chunk-transfer mechanics (5 MB–64 MB chunks, ≤128 MB final chunk, ≤1000 chunks, whole-upload if <5 MB); these are transfer-layer rules, not file-conformance rules, but are worth surfacing when the API profile is selected.
- All specification numbers herein reflect pages accessed 9 August 2026; TikTok changes these without notice and sometimes without changing visible "Last updated" stamps.